

API - Options Buying Strategy on Nifty 50 & Bank Nifty

Strategy Name:	Nifty 50 Momentum Options Buying Strategy
Version:	0.3.0
Segment:	Options (Nifty 50 & Bank Nifty Weekly/Monthly)
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User Guidance A — Strategy Overview

1. Objective of the Strategy

To systematically buy options (long premium only) on Nifty 50 and Bank Nifty indices using a multi-timeframe momentum-based signal engine. The strategy aims to capture directional moves driven by momentum bursts (T1 scalp: 1-15 min holds), intraday swing continuations (T2: 30 min to few hours), and positional multi-session trends (T3: hours to 1-3 days). The core philosophy is **defined risk, positive expectancy, and never selling/writing options**.

2. Type of Strategy Execution

EXECUTION The strategy uses **execution-style order placement** — signals are generated on candle closes and orders are placed immediately with smart order routing. It is *not* TWAP or VWAP based; it is event-driven and signal-triggered.

3. Operating System

Windows 10/11 (development) with production deployment on **Linux (Ubuntu 22.04)** via Docker containers on AWS EC2 (ap-south-1, Mumbai).

4. Data Used for Order Placement

The strategy places orders based on **both (i) tick-by-tick data and (ii) non-tick-by-tick data**:

- Tick-by-tick data:** Live WebSocket ticks from brokers (Zerodha Kite, FlatTrade, MStock) at up to 5,000 ticks/second. Used for real-time price monitoring, VWAP calculation, and bid/ask spread validation before entry.
- Non-tick (candle) data:** 1-minute, 5-minute, 15-minute, 30-minute, and 1-hour OHLCV candles aggregated from ticks. Signal generation **only fires on confirmed candle closes** to avoid whipsaws.

User Guidance B — Strategy Logic

1. How the Program Generates Signals and Executes Orders

Signal Generation Pipeline:

- 1. **Data Ingestion (Layer 1):** WebSocket streams pull real-time ticks from broker APIs. Ticks are normalized and published to Kafka topics (tick).
- 2. **Candle Building (Layer 2):** Ticks are aggregated into 1-minute OHLCV candles in-memory. Higher timeframes (5m, 15m, 30m, 1h, 1d) are derived from the 1m base.
- 3. **Storage (Layer 3):** Candles stored in TimescaleDB (writes) with Redis caching (reads). Kafka ensures at-least-once delivery.
- 4. **Analysis (Layer 4):** Technical indicators computed: RSI(14), EMA(9,21,50,200), MACD(12,26,9), ATR(14), Bollinger Bands(20,2), VWAP, ADX(14), Stochastic(14,3,3), OBV, Supertrend(10,3).
- 5. **Aggregation (Layer 5):** Market breadth computed — advance/decline ratio, percentage of stocks above VWAP, sector-wise strength.
- 6. **Regime Engine (Layer 6):** Multi-timeframe classifier (5m/15m/1h/Daily + breadth + VIX) publishes RegimeState with trend direction, strength (0-1), volatility tier, and phase.
- 7. **Signal Generation (Layer 6):** Adaptive strategy framework evaluates two plugin strategies:
 - **Momentum-Burst (T1):** 5m expansion candle + breadth confirmation + RSI-14 crossing ~55 + volume ≥ 1.2x avg + structure break above swing high/VWAP
 - **Trend-Pullback (T2):** Price near EMA21 in trending regime + RSI/volume rejection confirmationComposite score formula:
Score = (TrendScore × 0.40) + (MomentumScore × 0.30) + (BreadthScore × 0.30)
Signals fire when Score > 0.3 (Bullish) or < -0.3 (Bearish).
- 8. **Risk Gate (Layer 10):** Before entry — checks max concurrent positions (1), max trades/day (5), daily loss circuit breaker (≤ ₹2,500), time window (09:20-15:00 IST), instrument liquidity gate.
- 9. **Strike Selector:** ATM or ITM-1 strike, current or next weekly expiry, bid-ask spread ≤ maxSpreadPct.
- 10. **Order Placement:** Marketable limit order (LIMIT at ask + 0.5-1pt, IOC) placed via FlatTrade NorenApi. On fill, resting SL-M order placed immediately. On SL-M failure → MARKET exit (never naked).
- 11. **Monitoring:** Position tracked via broker order poll (every 2-3s). Trailing SL ratcheted after +12% profit. Hard stop at -18% premium (scalp) or -35% (positional). Time-stop if stalled ~10 min. Forced square-off at 15:15 IST.
- 12. **Journaling:** Every trade logged asynchronously to database for backtest/analysis.

2. Complete Entry and Exit Parameters

Entry Parameters — Momentum-Burst (T1 Scalp)

Parameter	Condition
Regime	BULLISH (A/D > 1.5, ≥2 top sectors UP, %aboveVWAP rising)
Momentum burst	5m candle range ≥ 1.5× ATR-14, close in top 25%, volume ≥ 1.2× 20-candle avg
Structure break	Close above prior swing high AND opening-range high AND VWAP
RSI confirmation	RSI-14(5m) crossing up through ~55

MACD	MACD histogram > 0 and expanding
Not extended	Price not > 2× ATR above VWAP
Volatility	India VIX in tradeable band, option bid-ask ≤ maxSpreadPct
Time window	09:20 - 15:00 IST

Exit Parameters

Parameter	T1 Scalp	T2 Intraday	T3 Positional
Hard stop loss	-18% of premium	-30% of premium	-35% of premium
Profit target	+25-40% of premium	+50-100% of premium	2R-3R on index
Trailing stop	Ratchet SL after +12%, 6% steps	Trail under swing structure	Trail under higher-lows
Time stop	~10 min stall	30 min stall	End of day (15:15)
Square-off time	15:15 IST	15:15 IST	15:15 IST (or next day if regime valid)

User Guidance C — Flow Chart of Strategy Logic and Order Execution

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+-----+ | START: Market Open | | (09:15 IST) | +-----+ | v +-----+
+-----+ | Tick Ingestion | | (WebSocket Stream) | +-----+ | v +-----+
+-----+ | Candle Formation | | (1m/5m/15m OHLCV) | +-----+ | v +-----+
| Regime Engine | | (Multi-TF Classify)| +-----+ | v +-----+ |
Strategy Selector | | (T1/T2/T3 based on | | Regime + Volatility| +-----+ | +-----+
+-----+ | | v v +-----+ +-----+ |Momentum | | Trend | |Burst | |Pullback| |Plugin | |Plugin | +---
+-----+ +-----+ | | +-----+ | v +-----+ | Composite Score | | > 0.3 / < -0.3?
| +-----+ | (Yes)| v +-----+ | Risk Gate Check | | - Max pos (1) | |
- Max trades (5) | | - Daily loss (Rs2500)| | - Time check (9:20- | | 15:00) | | - Liquidity check |
+-----+ | (Pass)| v +-----+ | Strike Selector | | (ATM/ITM-1, Weekly,
| | Spread ≤ limit) | +-----+ | v +-----+ | Order Placement | |
(Marketable Limit | | IOC at ask+0.5pt) | +-----+ | (Fill)| v +-----+
| SL-M Placement | | (Immediate resting) | +-----+ | +-----+ | | v v +-----+
+-----+ | SL-M | | SL-M | | SUCCESS| | FAILED | +-----+ +-----+ | | (Emergency) | v | +---
+-----+ | | MARKET | | | Exit | | +-----+ | v +-----+ | Position Monitoring |
| - Broker poll every 2-3s | | - Trailing SL after +12% | | - Time-stop check | +-----+
+-----+ | +-----+ | | | v v v +-----+ +-----+ +-----+ | Stop | |Target| | Time | |
Loss | | Hit | | Stop | +-----+ +-----+ +-----+ | | | +-----+ | v +-----+
+-----+ | Exit Order (MARKET | | or SL-M) | +-----+ | v +-----+ |
Journal Trade | | (DB write - async) | +-----+ | v +-----+ | END:
Ready for next| | signal | +-----+

```

Order Execution Flow Summary

1. **Signal fires** → Risk gate validates (concurrent positions, daily loss breaker, time window, liquidity)
2. **Strike selected** → ATM or ITM-1, nearest weekly/monthly expiry with adequate liquidity
3. **Entry order placed** → LIMIT IOC at ask + 0.5-1pt (marketable) via FlatTrade NorenApi
4. **On fill** → Resting SL-M order placed immediately at broker (never cancel-replace pattern)
5. **If SL placement fails** → MARKET order exits position immediately (never leave position unprotected)
6. **Position monitored** → Broker poll every 2-3s, trailing stop ratcheted after +12% profit
7. **Exit triggers** → Hard SL hit / Target reached / Time-stop (10min stall) / Forced square-off at 15:15 / Telegram /kill command
8. **Trade journaled** → All trade data written asynchronously to database for analysis

Risk Plan

Overall Risk Philosophy: Long premium only (defined risk). Maximum loss per trade is capped at the premium paid. No short options, no naked positions, no margin-intensive strategies. Daily loss circuit breaker at ₹2,500.

1. If your program is not working as per strategy?

- **Immediate action:** The program has a **health monitoring layer** that continuously validates signal parameters, order placement, and risk gate compliance. If any discrepancy is detected (e.g., signal fired outside allowed time, position opened without matching signal criteria), the **kill switch** is automatically triggered via Redis (execution:kill_switch=1).
- **Manual override:** Telegram /kill command immediately exits all open positions at MARKET price and prevents any new entries.
- **Audit trail:** Every decision (signal generation, order placement, order fills, exits) is logged with timestamps. Drift detection compares expected strategy action vs actual execution.
- **Recovery:** Strategy must be manually restarted after root cause analysis. Automated re-entry is blocked until human operator confirms fix.

2. If the loss is growing?

- **Per-position stop:** Hard stop loss of -18% (T1) to -35% (T3) of premium paid. These are **resting SL-M orders** placed at the broker the moment entry fills — they execute even if the program crashes.
- **Daily circuit breaker:** Maximum daily loss of ₹2,500 across all positions. Once hit, Redis kill switch flips to 1 — no new entries allowed for the rest of the day. The circuit breaker also alerts via Telegram.
- **Time stop:** If a position shows no movement in the expected direction within ~10 min (T1) or 30 min (T2), it is exited via MARKET order.
- **Position sizing:** Initial sizing at 2-3 lots (₹5,000-10,000 risk per trade), ensuring no single trade exceeds 1.5% of total capital.
- **Trailing stop:** After +12% profit, the stop is ratcheted up in 6% increments to lock in gains.

3. If Flattrade RMS squares off positions due to margin call?

- **Since the strategy buys long premium only, margin requirements are minimal** (full premium paid) — RMS square-off due to margin shortfall is unlikely for this strategy type.
- **If RMS square-off occurs anyway:**
 - The program detects the event via the broker reconciliation loop (polls broker order book every 2-3 seconds).
 - When it detects that the position no longer exists in the broker's position book but the internal state shows an open position, it logs the discrepancy and **immediately updates internal state** to reflect the square-off.
 - No further orders are placed for that particular signal.

- The daily loss circuit breaker total is updated to include the RMS square-off loss.
- An alert is sent via Telegram: ⚠ RMS SQUARE-OFF DETECTED for [symbol]

4. If the program crashes during market hours?

- **Resting SL-M orders protect open positions:** Since SL-M orders are placed at the broker and stored server-side by Flattrade, they remain active even if the program crashes. Positions will be automatically exited if stop loss is hit.
- **Broker reconciliation on restart:** When the program restarts, it queries the broker's position book and order book to re-sync state. Any positions that were opened by the program but not tracked locally are detected and properly monitored.
- **Forced square-off at 15:15:** If the program remains down until market close, the broker's RMS square-off will handle any remaining positions (since long options expire worthless or have value).
- **PM2 process manager:** The API server runs under PM2 with auto-restart on crash (ecosystem.config.js). In production (Docker), Docker's restart: unless-stopped policy handles automatic recovery.
- **Telegram notification:** The health monitoring system sends an alert if the program heartbeat is not received for more than 30 seconds.
- **Backup plan:** In an emergency, user can log into the Flattrade web/mobile dashboard directly and manually exit any positions. The Telegram /kill command also works if the backend API is still running.

Declaration

I hereby confirm that the information provided in this document is true and accurate to the best of my knowledge. I understand that algorithmic trading involves financial risk and I take full responsibility for the strategy described above.

Date: ____/____/____

Place: Gurgaon, Haryana

Signature:

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Saral Jyotish Trading System • For Flattrade API Submission • Document generated on ' + (Get-Date -Format 'dd/MM/yyyy HH:mm') + '@'